

SMART AGRICULTURE SYSTEM WITH EDGE COMPUTING AND DATA ANALYSIS

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Edge Hardware Setup - Overview

Role of Edge Device in Smart Agriculture

- Collect real-time environmental metrics: temperature, humidity, and light intensity.
- Execute local feedback logic for fast responses without waiting for cloud processing.
- Display status and alerts directly on local indicators (LCD/OLED and RGB LED).
- Bridge sensor layer with cloud services through MQTT on Adafruit IO.

Practical impact

- Reduce latency for safety alerts.
- Maintain stable operation under intermittent connectivity.

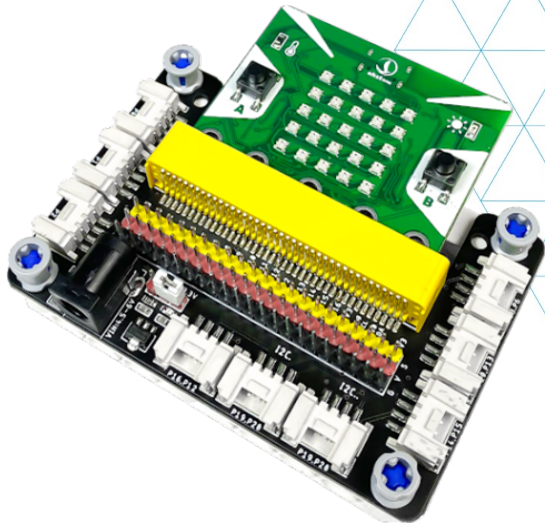
Edge Hardware Setup - Core Architecture

Core blocks

- Yolo:Bit (ESP32 dual-core) as edge controller.
- Extend Shield for I/O, I2C, and SPI expansion.
- Wi-Fi link for cloud communication via MQTT.
- Stable power by adapter/battery.

Why this setup

- Fast prototyping, easy wiring, simple maintenance.



Yolo:Bit + Extend Shield hardware block ▶

Edge Hardware Setup - Sensors & Actuators

Sensor Inputs

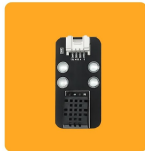
- **DHT20 (I2C)**: temperature and humidity sensing.
- **LDR (ADC)**: light intensity monitoring.
- **Infrared sensor**: intrusion/motion detection.

Design objective: close the loop between sensing, alerting, and actuation at edge level.

Indicators & Actuation

- **LCD/OLED**: local status and warning messages.
- **RGB LED**: green (stable), red (high temperature/alert).
- **Fan / Pump / Servo**: cloud-assisted environmental control.

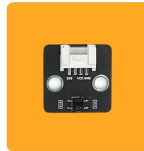
HW Sensors & Detail Images



Temp/Humidity



Light



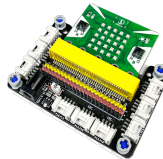
Infrared



LCD detail

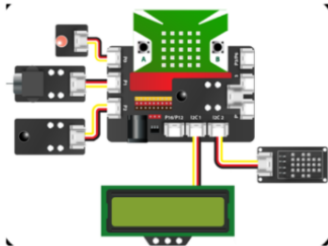


RGB detail

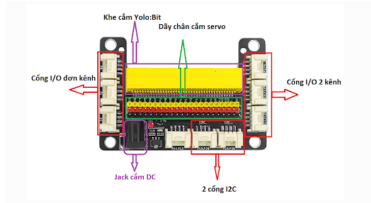


Yolo:Bit detail

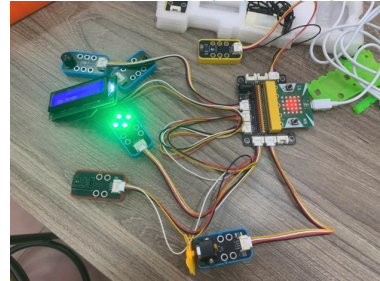
Edge Hardware Setup - HW Images



HW 1



HW 2



HW 3

Edge Hardware Setup - Data Flow (Edge to Cloud)

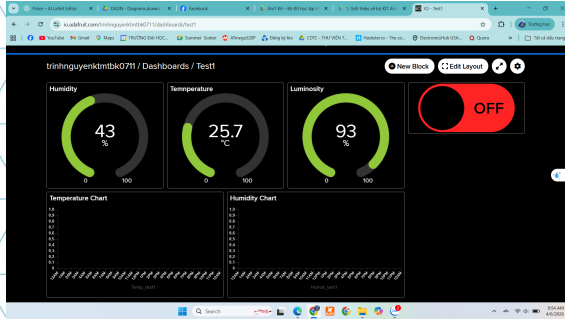
Operational pipeline

- ➊ **Setup phase:** initialize sensors, connect Wi-Fi, authenticate MQTT client.
- ➋ **Main loop (every 5s):** sample DHT20/LDR and apply local threshold logic.
- ➌ **Publishing:** push telemetry to Adafruit IO feeds.
- ➍ **Subscription:** receive actuator commands from backend services.

MQTT feeds (example)

- Telemetry: temp-test1, humid-test1, luminor-test1
- Control: actuator-fan, actuator-pump, actuator-led

Edge Hardware Setup - Dashboard & Feed



Dashboard overview

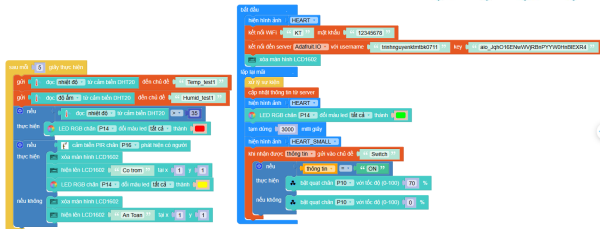
☐ Humid_test1	humid-test1	43.0	18 days ago	
☐ Luminor_test1	luminor-test1	93	18 days ago	
☐ Switch	switch	OFF	18 days ago	
☐ Temp_test1	temp-test1	25.7	18 days ago	

Feed monitoring interface

Edge Hardware Setup - Integration Highlights

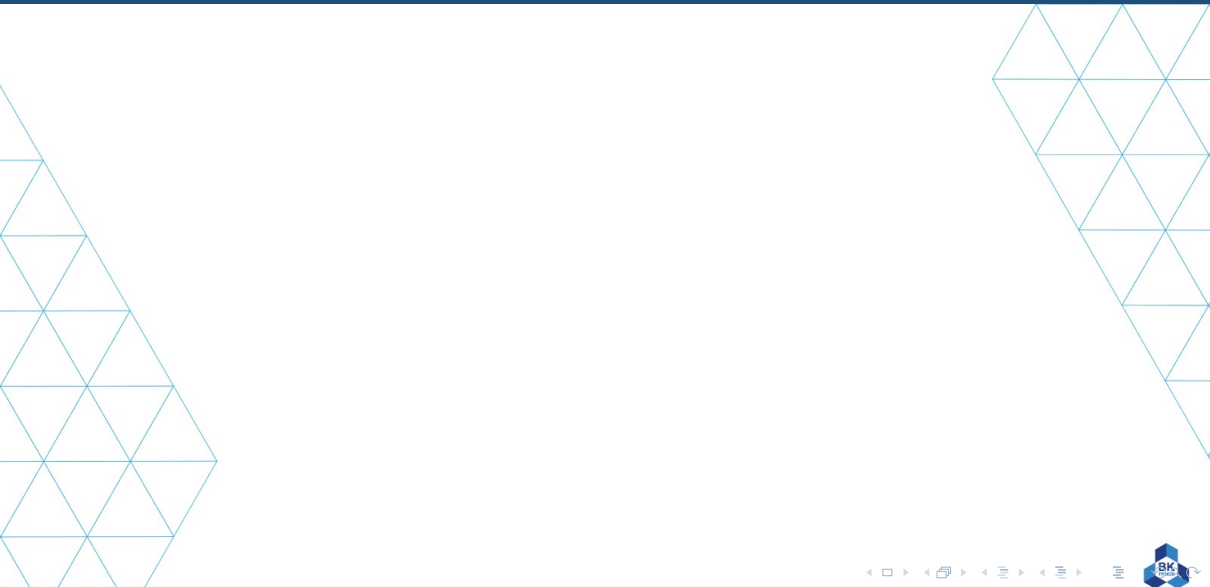
Key highlights

- MQTT + Adafruit IO for lightweight, stable telemetry.
- Local rule: high temperature triggers LCD + red LED alert.
- IR events are alerted locally before cloud confirmation.
- Reliable edge-to-cloud flow for dashboard automation.



Edge code stream (OhStem)

Load Balancer Architecture



Load Balancer Configuration

Configuration

- Implemented as **custom Node.js middleware**
- Acts as the **central traffic controller**
- Configuration controlled by:
upstream.conf
- Enables:
 - Flexible deployment
 - Easy scaling
 - Maintainable architecture

Subscriber Initialization

- Reads **adafruit_io** configuration
- Establishes:
 - Persistent MQTT connection
 - Authenticated session
- Subscribes to sensor topics:
 - Temperature
 - Humidity
- Converts received data into:
Standard JSON format

upstream.conf Structure

Configuration file divided into 3 main blocks:

- **http_port**
 - Defines listening port
 - Example: 8080
- **backends**
 - List of backend servers
 - Used for load distribution
- **adafruit_io**
 - MQTT connection settings
 - Topic subscriptions

```
{  
  "http_port": 8080,  
  "backends": [{  
    "host": "127.0.0.1",  
    "port": 3001  
  }, {  
    "host": "127.0.0.1",  
    "port": 3002  
  }],  
  "adafruit_io": {  
    "host": "mqtt://io.adafruit.com",  
    "username": "...",  
    "key": "...",  
    "topics": [  
      "sensor.temperature",  
      "sensor.humidity"  
    ]  
  }  
}
```

HTTP Router and Round-Robin:

- Reads backend list from: **backends**
- Uses: **Round-Robin Scheduling**
- Distributes requests sequentially:
 - Backend 1
 - Backend 2
 - Backend 3
- Handles:
 - External client requests
 - Internal MQTT data

System Benefits:

- Even load distribution
- Reduced risk of:
 - Server overload
 - Single-point failure
- Improved:
 - Scalability
 - Fault tolerance
 - High availability
- Easy backend expansion

Data Visualization

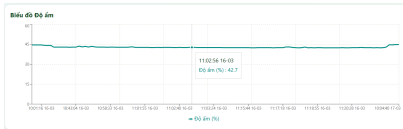


Figure 1: Humidity



Figure 2: Light

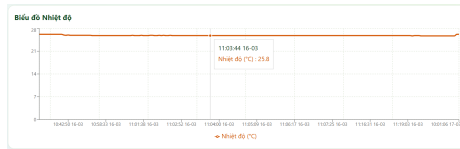


Figure 3: Temperature

- Interactive charts provide real-time and historical views of environmental metrics
- Each chart supports dynamic updates for better monitoring and analysis
- Help users quickly understand trends and current system conditions

Filtering & Aggregation

NÔNG NGHIỆP THÔNG MINH

Dashboard IoT Nông Trại

Theo dõi ánh sáng, nhiệt độ, độ ẩm theo thời gian và nhận cảnh báo sớm.

Làm mới ngay

Tự động cập nhật mỗi 5 phút
Lần gần nhất: 22:47:35 3/4/2026

Figure 4: Data refresh process

Filtering interface showing options for data range and export.

Khoảng lọc: Mặc định - chi tiết Từ ngày: mm/dd/yyyy Đến ngày: mm/dd/yyyy Xuất CSV

Figure 5: Filtering interface

- **Data Refresh**

- Manual refresh by user
- Auto update

- **Time Filtering**

- Day / Week / Month views
- Custom date range selection

- **Aggregation**

- Raw data for detail inspection
- Average values for trend analysis

- **Benefits**

- Reduce noise in sensor data
- Highlight long-term patterns

Status Evaluation & Alerts

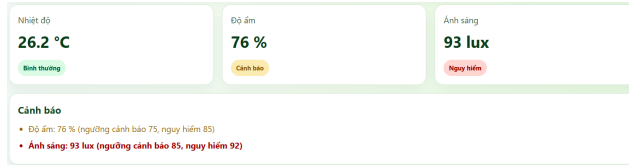


Figure 6: System alert visualization

- **System States:**

- Normal: within safe range
- Warning: approaching critical limits
- Critical: exceeds safe thresholds

- **Threshold Comparison:**

- Evaluate data in ideal and critical ranges

- **Alert Mechanism:**

- Dashboard notifications
- On-site RGB LED and LCD warnings

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- THE END -

Thank you for your attention

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